



Lesson 2 – Sensing the World: From Signals to Data

Teacher-only slide – Lesson Overview

Lesson Preparation



Before you start:

- Print of the Worksheet “Encode a Mystery Word in Binary” for every student
- Familiarize yourself with the slides and activities in this lesson.
- You can print off the teacher slides or add any notes/ parts of the example scripts to the notes on the student slides. This way, you can see them when in presenter mode.
 - Then hide the teacher slides, so they don’t get shown when you present.
- We recommend to go through the slides in presenting mode, so that you also become familiar with the animations and order in which the different items appear.

Encode a Mystery Word in Binary

Choose a word that has between 4 and 8 characters. Use the letter-to-binary code below to translate your word to binary.

Letter	Binary	Squares
A	00001	☐ ☐ ☐ ☐ ☐
B	00010	☐ ☐ ☐ ☐ ☐
C	00011	☐ ☐ ☐ ☐ ☐
D	00100	☐ ☐ ☐ ☐ ☐
E	00101	☐ ☐ ☐ ☐ ☐
F	00110	☐ ☐ ☐ ☐ ☐
G	00111	☐ ☐ ☐ ☐ ☐
H	01000	☐ ☐ ☐ ☐ ☐
I	01001	☐ ☐ ☐ ☐ ☐
J	01010	☐ ☐ ☐ ☐ ☐
K	01011	☐ ☐ ☐ ☐ ☐
L	01100	☐ ☐ ☐ ☐ ☐
M	01101	☐ ☐ ☐ ☐ ☐
N	01110	☐ ☐ ☐ ☐ ☐
O	01111	☐ ☐ ☐ ☐ ☐
P	10000	☐ ☐ ☐ ☐ ☐
Q	10001	☐ ☐ ☐ ☐ ☐
R	10010	☐ ☐ ☐ ☐ ☐
S	10011	☐ ☐ ☐ ☐ ☐
T	10100	☐ ☐ ☐ ☐ ☐
U	10101	☐ ☐ ☐ ☐ ☐
V	10110	☐ ☐ ☐ ☐ ☐
W	10111	☐ ☐ ☐ ☐ ☐
X	11000	☐ ☐ ☐ ☐ ☐
Y	11001	☐ ☐ ☐ ☐ ☐
Z	11010	☐ ☐ ☐ ☐ ☐

Fill the 'bit boxes' with the corresponding zeros and ones (0 and 1).
Leave the lines below the boxes blank for your classmate to fill in.

☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐
_____	_____	_____	_____
☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐	☐ ☐ ☐ ☐ ☐
_____	_____	_____	_____

Teacher-only slide – Lesson Overview

Lesson 2 - Overview



1. **Lesson 1 Recap** - 5 min
2. **Sensors** – 15 min
 - 2.1 From Human to Robot Senses
 - 2.2 Types of Sensors
3. **Language of Machines** – 30 min
 - 3.1 Control Process in Humans and Robots
 - 3.2 Electricity (off/on)
 - 3.3 Bits and Bytes
 - 3.4 (Spelling in) Binary
 - 3.5 Optional extension: Binary beads or Counting in binary – 15-30 min
 - 3.6 Programming Languages
4. **Building Blossom's head** - 30 min
5. **Survey** - 5 min
6. **Summary** - 5 min

Total: 90 min

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Lesson 2 – Learning objectives



After this lesson, students will be able to:

- explain the function of sensors in robots and other machines,
- identify different types of sensors and what each one senses,
- describe the control process sequence in robots and other machines (sensor input → computer with code → motor → movement)
- describe how computers use 0/1 (off/on) to represent information,
- encode and decode a short word using a binary code chart,
- understand what programming languages are.

Teacher-only slide – Lesson Overview

Lesson 2 - Student success criteria



- I can explain what sensors are used for inside of robots and other machines.
- I can name three different types of sensors and what they measure.
- I can outline the order of processing inside the Blossom robot that is needed to turn environmental input into an action. (input → sensor → computer with code → motor position → movement)
- I can describe how computers use electricity to communicate information with off- and on-states (0/1).
- I can encode and decode a short word in binary using the class chart.
- I can explain that programming languages are a human-friendly way to write instructions that computers can understand.

Teacher-only slide – Instruction Guide

Lesson 1 Recap



- **Robot types** (selection): industrial, medical, entertainment, discovery, educational, domestic, social, humanoid
- **What is a robot?:** A machine that senses (uses sensors to notice things in the world around it), moves (has moving parts that can act in the real world) and is intelligent (uses code to make a decision about the next action from the input)
- **Robot parts:** motor (movement), sensor (notice), computer + software/code (control/process), wires (connect), power source (energy), machinal structure (hold everything together)
- **Helpful uses:** dangerous, dull, precise, labour shortage
Harmful uses: replacing humans, befriending humans (emotional dependency, illusion of care but don't really understand, social isolation, inappropriate responses)

Let's remember what we learned in Lesson 1!



- 3 robot examples (types or jobs)
- What is a robot?
- 3 robot parts & what they do
- 2 helpful + 2 harmful uses of robots

Teacher-only slide – Instruction Guide

Sensors Intro



Example Script:

- *“You learned in the previous lesson that robots (and other technologies) use sensors to notice what is happening around them. Today we will zoom into this aspect a bit more to identify how sensors can turn information from the world around them into a language that robots (or other computers) can understand.”*
- **Whole-class brainstorm: How do we sense the world?**

Review the human senses and what the “biological sensors” notice around us, i.e. the sensory organs: eyes (vision), ears (sound), nose (smell), tongue (taste), skin (touch).
- Then ask the students: *“Which of these senses do you think a machine could also have? Are there any senses a robot can have that we don’t?”*

Human Senses



- help us notice things in our environment



Which of these senses
could a robot also have?

Teacher-only slide – Instruction Guide

Robots Senses



Example Script:

- *“Similar to how humans have sensory organs to notice things around them, robots can use sensors to measure different information in their environment.”*
- *“Almost everything that can be measured or captured with a machine could also be used as a sensor for robots. This allows them to sense many things around them. Some robot senses are very similar to human senses, while others can be quite different in terms of the information they receive and how sensitive they are. We will look at some examples now.”*

Machine Sensors



- help them notice things in the environment



SIGHT



HEARING



TOUCH



Teacher-only slide – Instruction Guide

Robots Senses



Example Script:

- *“There are many other sensors that can be used inside of a robot to give it the ability to sense further information from its environment”*

Instruction Note:

- Introduce the sensors on the next slide to give students a sense of what kinds of things sensors can measure.
 - **vision/imaging sensors** (similar to our sense of vision): camera, colour sensor (can detect different colours), light sensor (can measure how bright or dark the robot’s surroundings are), ultrasonic sensor (senses how far the next object is away)
 - Highlight that the ultrasonic sensor (can also be called distance sensor) is the sensor that is used in Blossom now.
 - **tactile sensors** (similar to our sense of touch): button (the simplest type of sensor, only detects if it being pushed or not), touch sensor (similar to a laptop’s touchpad)
 - **other sensors**: sound sensors/microphones, temperature sensor
- The sensors on the next slide could all be used inside of Blossom after adjusting the code on its computer accordingly.

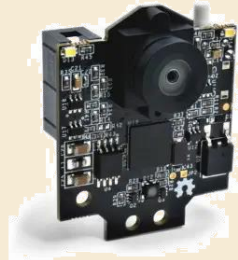
Sensor Types



vision & imaging sensors



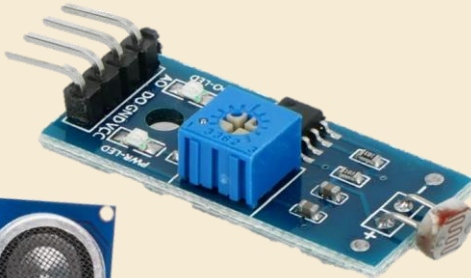
colour sensor



camera

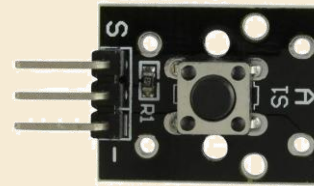


distance sensor
(ultrasonic sensor)



light sensor

tactile sensors

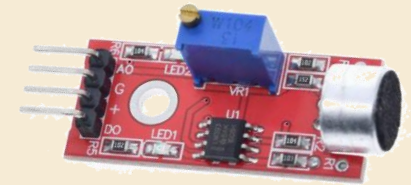


button

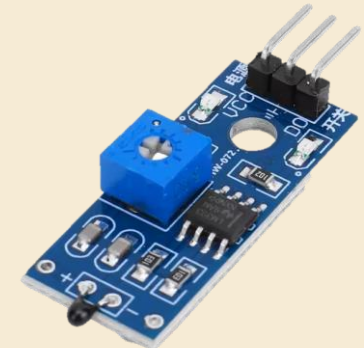


touch sensor

other sensors



sound sensor
(microphone)



temperature sensor

Teacher-only slide – Instruction Guide

Sensor Types



Instruction Note:

On the next slide, show one robot picture at a time and let students think about which kinds of sensors the depicted robots have and what they sense in the environment.

Note: Some robots (like Nao) have lots of sensors. We have provided a full list for your reference, but you don't need to work through all examples. We suggest getting students to identify at least 3 .

Bee bot:

- buttons (students can tell the bot what to do)
- newer versions also come with a microphone (to detect sound) and infrared sensors (to detect other bee bots)



Bee Bot

Teacher-only slide – Instruction Guide

Sensor Types



- Vending machine:

If it is difficult for the students to think of the sensors in a vending machine, you can go through the whole process of buying something from a vending machine and ask them to consider what the machine needs to know to select the correct actions throughout the process):

- **buttons:** input of what item user selects
- **temperature sensor:** determine how much the food/drinks need to be cooled
- **coin mechanism** (combination of optical, mechanical and electromagnetic sensors): identify coin size, weight and metallic composition
 - additionally, some vending machines have more complex sensors for processing cashless payments
- can have **security sensors** such as camera, tilt sensor (alarm when it falls over), or motion detector

Teacher-only slide – Instruction Guide

Sensor Types



- **NAO** (football playing robot) has the following sensors:
 - chest **button** -> processing simple commands, f.i. to turn on and off,
 - **camera** -> images of surrounding, in combination with additional code it can be used to identify objects in the robot's surrounding,
 - audio/**microphone** -> interacting with users through spoken language,
 - ultrasonic/**proximity sensor** -> to sense how far away objects/people are,
 - **touch sensor** on head -> can be programmed to complete certain action when people pet its head,
 - **temperature**/thermal sensor -> can block motors when they are overheating
 - **motor position sensors** for every motor -> knows which position it is in (do not mention unless you want to introduce this type of sensor)
 - **gyroscope sensor** -> knows when it's upright and when it falls (do not mention unless you want to introduce this type of sensor)



NAO robot

Teacher-only slide – Content Extension

Sensor Types



- There are a lot of different types of sensors that can be used for robots (or other machines) to detect certain information in their surroundings.
- Some other examples that were not mentioned on the student slides are:
 - **position sensor:** e.g. to know the position of a motor (Blossom's motors cannot do this but the NAO robot in the photo has these sensors), or fluid level sensors
 - **motion sensor:** e.g. tilt sensor (detects tilting of object/device), accelerometer (measuring acceleration), or gyroscope sensor (uses gravity and momentum forces to measure rotation, a robot knows if it's upright or when it falls)
 - **more:** air pressure sensor, humidity sensor, heartbeat sensor, motion sensor (as used when lights turn on when people walk past), gas and chemical sensors (detecting different gases and chemicals, such as alcohol sensor, CO-sensors that set off fire alarms)
- Feel free to also introduce some of these to the students when discussing sensors.

Machine Sensors



What sensors could these robots have?
– and what do they sense?



Bee Bot



Vending
Machine



NAO robots

Teacher-only slide – Instruction Guide

Control Process inside Digital Systems



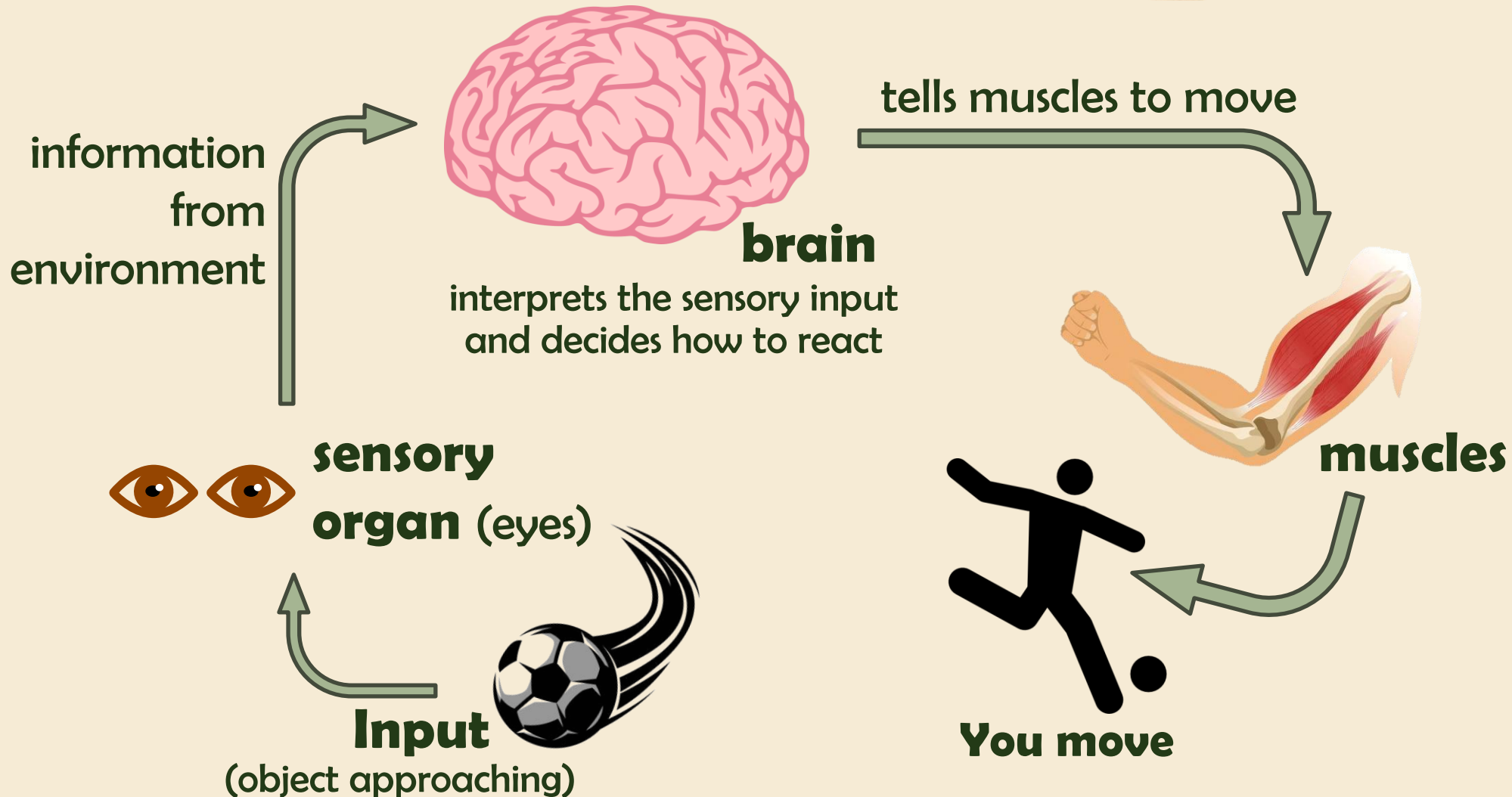
Example Script:

- *“Now that we know that both humans and robots use senses/sensors to make sense of the world around them, we will have a look at the control process. This is the sequence of steps needed to turn the environmental input into an action.”*

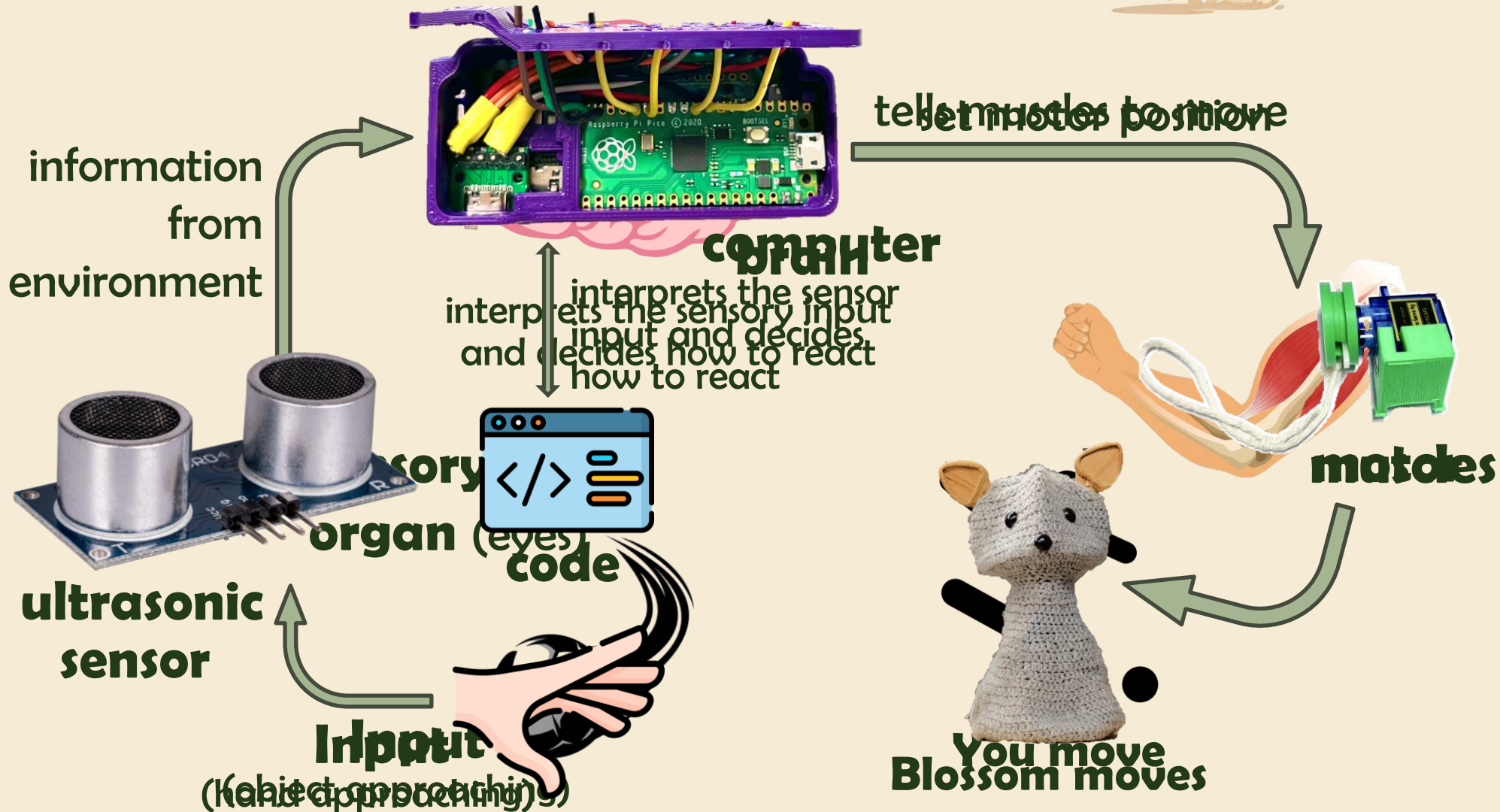
Instruction Note:

- On the next slide, the human control process from sensory input (a ball that is coming towards you) to the motor response (you moving to kick the ball) is visualized. Every item appears after each other.
- The slides following will one-by-one swap out a part of the visualization to show what the process looks like in the Blossom robot when a hand is approaching the ultrasonic sensor on its head.
- You can let students predict what the next swap will be (e.g. ask: *“What does Blossom have in place of a brain?”*, *“What fulfills the same function as a human muscle in a robot?”*)

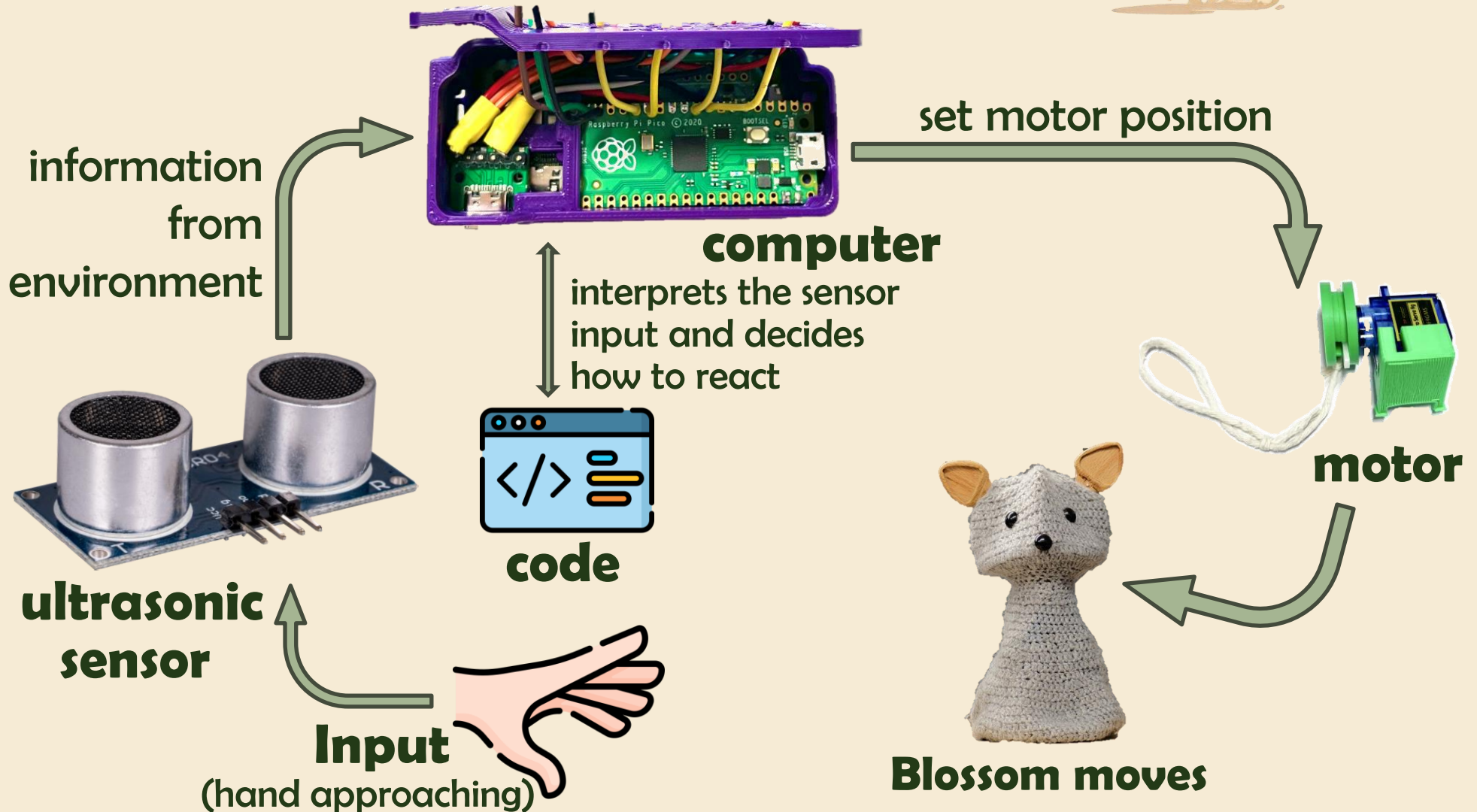
Human Control Process



Robot Control Process



Robot Control Process



Teacher-only slide – Instruction Guide

Binary

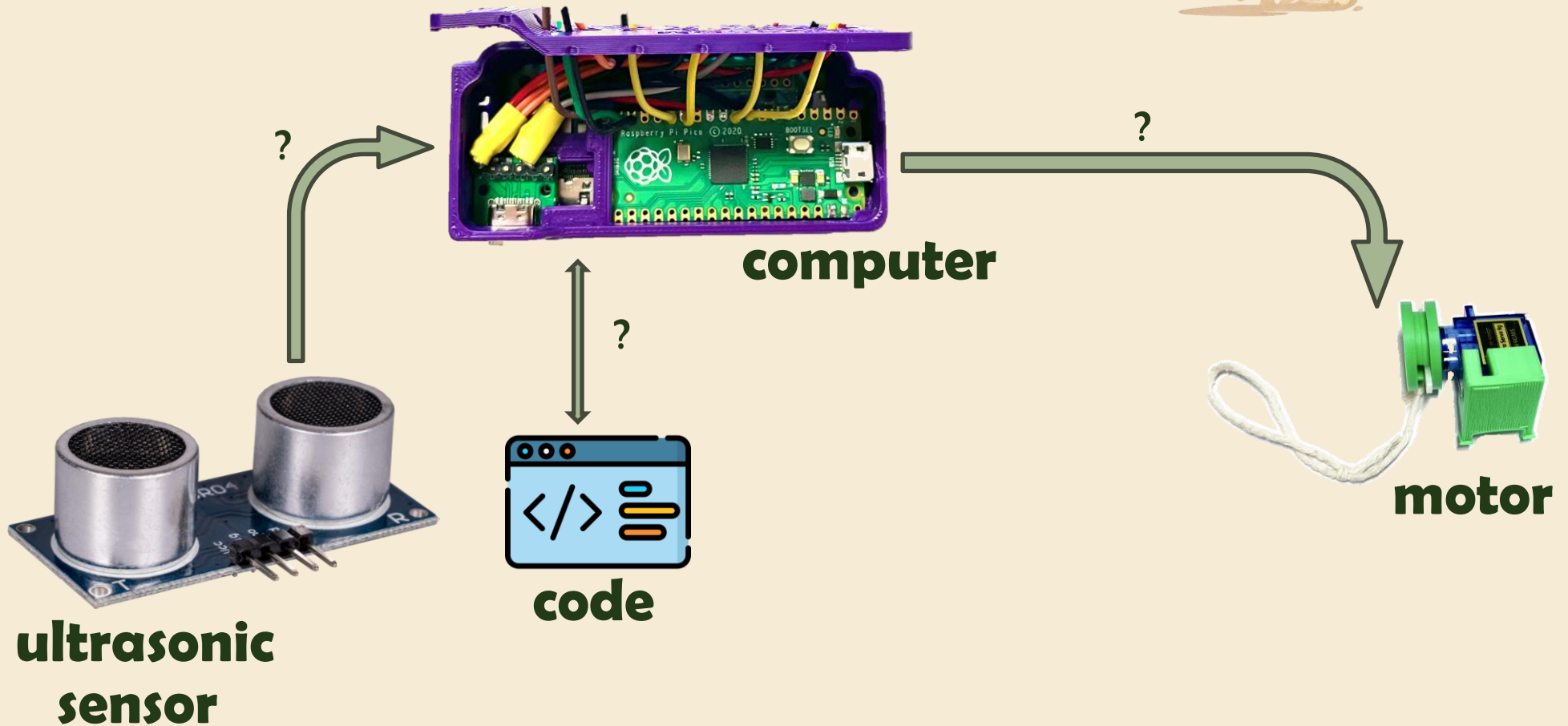


Example Script:

- *“Now we know that sensors help robots gather information (e.g., sound waves, light or press of a button). The different robot parts now need to convert this information into ‘messages’ that its computer can use to make decisions – such as how to respond or move through the environment.*
- *Computers are electrical, so all information that is sent into a computer, processed in a computer or sent out from computer is in the form of electrical signals.*
- *A simple electrical device we all know are light bulbs. They can be either on or off.*
- *Inside of computers, information is processed with lots of tiny micro transistors that work like lightbulbs. They can also either be flipped on or off, depending on whether they receive electrical input or not”*

Instruction Note:

- When talking about electrical signals being on or off, you can show this using the light switch of the classroom light.
- A great video explanation of this: <https://www.youtube.com/watch?v=Xpk67YzOn5w>



How do the parts communicate?

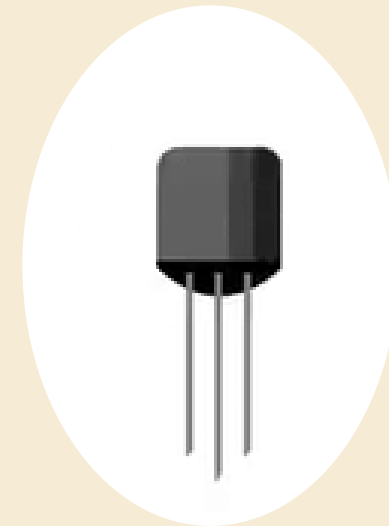
Binary



- Computers are electrical, all information is sent in electrical signals



The on- & off-state can be coded as numbers (zero and one)



micro transistor
in computers

Teacher-only slide – Instruction Guide

Binary Sequences



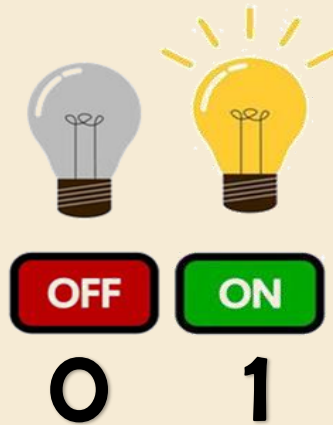
Instruction Note:

- Let students find the sequence of zeros and ones from the image of on and off transistors/lights. Every light that is on is a 1, every that is off is a 0.
- The numbers will appear after one another in the order of the sequence.

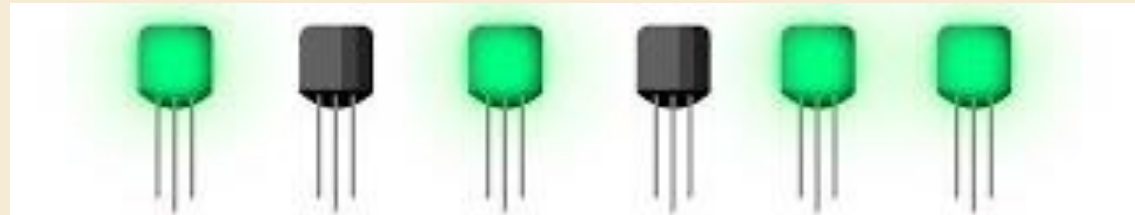
Binary Sequences



- Computers are electrical, all information is sent in electrical signals



1 0 1 0 1 1



Teacher-only slide – Instruction Guide

Bits and Bytes



Example Script:

- “Every digit (0 or 1) in binary is called a Bit (**binary** digit)
- *Since a single on/off of a digit cannot express much information, multiple of these bits are combined in computers to express more complex information.”*

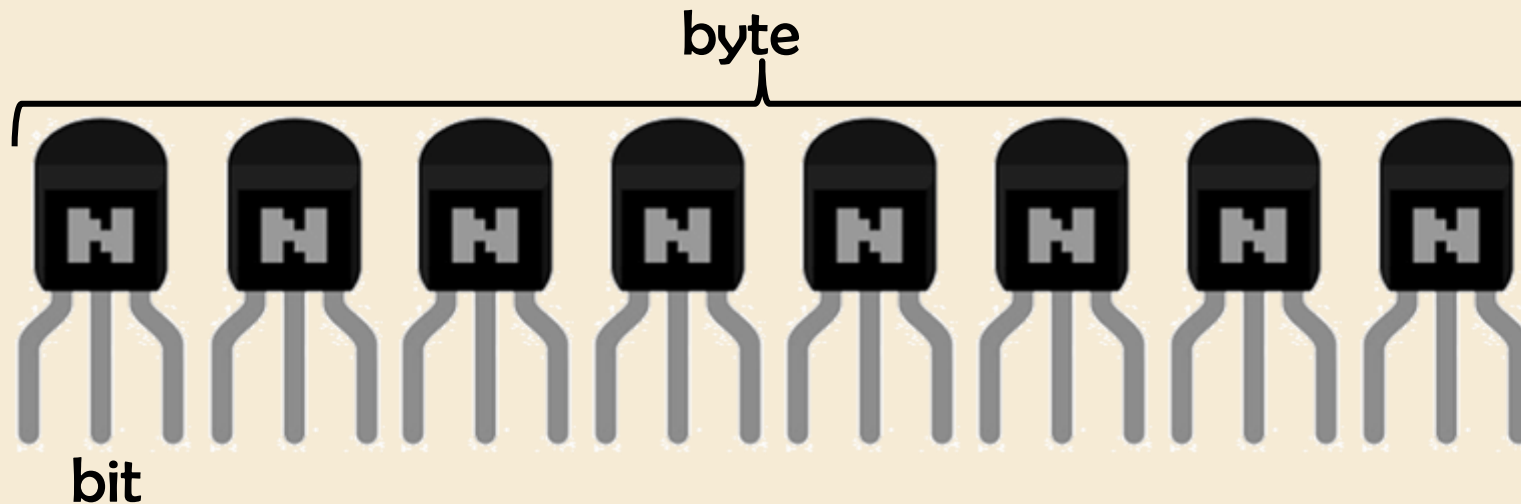
Instruction Note:

- Students might know this word already from MB/mega bytes (1 million bytes) or GB/giga bytes (1 billion bytes). This is used both when talking about storage (how much data is saved inside a computer’s memory) and related to data usage (how many bytes of data did the device download from or upload to the internet).

Bits and Bytes



- Every digit (0 or 1) is called a bit.
- In most computers, sequences of 8 bits are combined to store or process information
- A sequence of 8 bits = 1 byte



Spelling in Binary



- Every symbol you can type on a computer is stored as a byte.
- Example: “Hi!”

01001000

H

01101001

i

00100001

!

Teacher-only slide – Instruction Guide

Spelling in Binary Activity



Instruction Note:

- Pair the students with one other student (or a group of three, they will then exchange their writing in a circular way).
- Hand out the worksheet “Spelling in Binary”
- Tell them to:
 - “Think of a word you want to write to your partner (4-8 characters).”
 - “Use the worksheet and the binary code reference on the slide to encode it in binary.”
 - You can either write zeros and ones into the ‘bit boxes’, or colour in all that are on / a 1)”
 - Leave the line underneath the “bit boxes” blank, they are for your partner to decode your word.
- When both students are done, let them exchange their binary writing with the partner and try to decode what they wrote (also using the reference on the slide).

Note: While students are completing the worksheet, leave the “Spelling in Binary 101” slide up for them to easily refer to the code conversion chart.

Encode a Mystery Word in Binary

Choose a word that has between 4 and 8 characters. Use the letter-to-binary code below to translate your word to binary.

Letter	Binary	Squares	Letter	Binary	Squares	Letter	Binary	Squares	Letter	Binary	Squares
A	0001	☐ ☐ ☐ ☐	H	0100	☐ ☐ ☐ ☐	N	0110	☐ ☐ ☐ ☐	U	0101	☐ ☐ ☐ ☐
B	0010	☐ ☐ ☐ ☐	I	0001	☐ ☐ ☐ ☐	O	0111	☐ ☐ ☐ ☐	V	0110	☐ ☐ ☐ ☐
C	0011	☐ ☐ ☐ ☐	J	0010	☐ ☐ ☐ ☐	P	1000	☐ ☐ ☐ ☐	W	0111	☐ ☐ ☐ ☐
D	0010	☐ ☐ ☐ ☐	K	0011	☐ ☐ ☐ ☐	Q	1001	☐ ☐ ☐ ☐	X	1000	☐ ☐ ☐ ☐
E	0011	☐ ☐ ☐ ☐	L	0100	☐ ☐ ☐ ☐	R	1000	☐ ☐ ☐ ☐	Y	1001	☐ ☐ ☐ ☐
F	0010	☐ ☐ ☐ ☐	M	0101	☐ ☐ ☐ ☐	S	1001	☐ ☐ ☐ ☐	Z	1010	☐ ☐ ☐ ☐
G	0011	☐ ☐ ☐ ☐	X	0110	☐ ☐ ☐ ☐	T	1000	☐ ☐ ☐ ☐			

Fill the 'bit boxes' with the corresponding zeros and ones (0 and 1).
Leave the lines below the boxes blank for your classmate to fill in.

☐ ☐ ☐ ☐	☐ ☐ ☐ ☐	☐ ☐ ☐ ☐	☐ ☐ ☐ ☐
_____	_____	_____	_____
☐ ☐ ☐ ☐	☐ ☐ ☐ ☐	☐ ☐ ☐ ☐	☐ ☐ ☐ ☐
_____	_____	_____	_____

worksheet

Spelling in Binary 101

A	0 0 0 0 1	K	0 1 0 0 1	U	1 0 1 0 1
B	0 0 0 1 0	L	0 1 1 0 0	V	1 0 1 1 0
C	0 0 0 1 1	M	0 1 1 0 1	W	1 0 1 1 1
D	0 0 1 0 0	N	0 1 1 1 0	X	1 1 0 0 0
E	0 0 1 0 1	O	0 1 1 1 1	Y	1 1 0 0 1
F	0 0 1 1 0	P	1 0 0 0 0	Z	1 1 0 1 0
G	0 0 1 1 1	Q	1 0 0 0 1		
H	0 1 0 0 0	R	1 0 0 1 0		
I	0 1 0 0 1	S	1 0 0 1 1		
J	0 1 0 1 0	T	1 0 1 0 0		

slide with conversion chart

Teacher-only slide – Content Extension

Spelling in Binary Code



- The code on the next slide is a simplified version of the common ASCII code (American Standard Code for Information Interchange)
- ASCII normally uses 7 or 8 bits to encode a single character, whereas our simplified code only uses 5
- If you were to add '010' before the numbers in our simplified code, you would get the ASCII code version for uppercase letters.

If you added '011' in front, you would get the lowercase letters.

ASCII Char	Hex	Bin	ASCII Char	Hex	Bin		
65	A	41	0100 0001	97	a	61	0110 0001
66	B	42	0100 0010	98	b	62	0110 0010
67	C	43	0100 0011	99	c	63	0110 0011
68	D	44	0100 0100	100	d	64	0110 0100
69	E	45	0100 0101	101	e	65	0110 0101
70	F	46	0100 0110	102	f	66	0110 0110
71	G	47	0100 0111	103	g	67	0110 0111
72	H	48	0100 1000	104	h	68	0110 1000
73	I	49	0100 1001	105	i	69	0110 1001
74	J	4A	0100 1010	106	j	6A	0110 1010
75	K	4B	0100 1011	107	k	6B	0110 1011
76	L	4C	0100 1100	108	l	6C	0110 1100
77	M	4D	0100 1101	109	m	6D	0110 1101
78	N	4E	0100 1110	110	n	6E	0110 1110
79	O	4F	0100 1111	111	o	6F	0110 1111
80	P	50	0101 0000	112	p	70	0111 0000
81	Q	51	0101 0001	113	q	71	0111 0001
82	R	52	0101 0010	114	r	72	0111 0010
83	S	53	0101 0011	115	s	73	0111 0011
84	T	54	0101 0100	116	t	74	0111 0100
85	U	55	0101 0101	117	u	75	0111 0101
86	V	56	0101 0110	118	v	76	0111 0110
87	W	57	0101 0111	119	w	77	0111 0111
88	X	58	0101 1000	120	x	78	0111 1000
89	Y	59	0101 1001	121	y	79	0111 1001
90	Z	5A	0101 1010	122	z	7A	0111 1010
<linuxhint/>							

<linuxhint/>

Spelling in Binary

1 0 1



A 0 0 0 0 1

K 0 1 0 0 1

B 0 0 0 1 0

L 0 1 1 0 0

U 1 0 1 0 1

C 0 0 0 1 1

M 0 1 1 0 1

V 1 0 1 1 0

D 0 0 1 0 0

N 0 1 1 1 0

W 1 0 1 1 1

E 0 0 1 0 1

O 0 1 1 1 1

X 1 1 0 0 0

F 0 0 1 1 0

P 1 0 0 0 0

Y 1 1 0 0 1

G 0 0 1 1 1

Q 1 0 0 0 1

Z 1 1 0 1 0

H 0 1 0 0 0

R 1 0 0 1 0

I 0 1 0 0 1

S 1 0 0 1 1

J 0 1 0 1 0

T 1 0 1 0 0

Extension Activity



Binary Bracelets

- Start off as in the previously described activity. Instead of having a partner decode the word the students encoded, let them turn it into a bracelet with light and dark coloured beads
- You could still let a partner decode it first to check that there are no errors in the encoded bits.



Extension Activity: Counting in Binary

- You can extend this knowledge into maths class and introduce counting in binary
- In our normal counting system, each digit represents an increasing power of ten (decimal system), so that
$$3702 = 2 \times 10^0 + 0 \times 10^1 + 7 \times 10^2 + 3 \times 10^3$$
 - the digits can take on the values from 0-9
- In the binary system, each digit instead represents an increasing power of two. Each digit can only take on the value 0 or the value 1.
$$10010 = 1 * 2^1 + 1 * 2^4 \text{ (17 in decimal system)}$$
- A byte can represent any number from 0 up to 255.

Teacher-only slide – Instruction Guide

Translating between Computer and Human Language



Example Script to introduce programming languages:

- “As you can see in the last activity, writing in binary takes a lot more time and characters than writing in English – or other alphabetical languages”
- ‘We still want to be able to use computers and communicate what they should do, without having to write a ton of zeros and ones.’
- ‘For this, people have invented something called programming languages, which act as a kind of translator between human language and computer language.’
- ‘There are many different programming languages that have been invented. They contain **words from natural language** but also **tell the computer exactly what to do**. Some examples are Python, HTML, Java, C#, C++, JavaScript, SQL, R.’

Instruction Note:

- Students will get to work with a very simple version of a programming language (block-based coding) in the next lesson, and learn more about some basic concepts in coding (e.g. loops, if-statements).

Programming Languages



Have you heard of programming languages before?

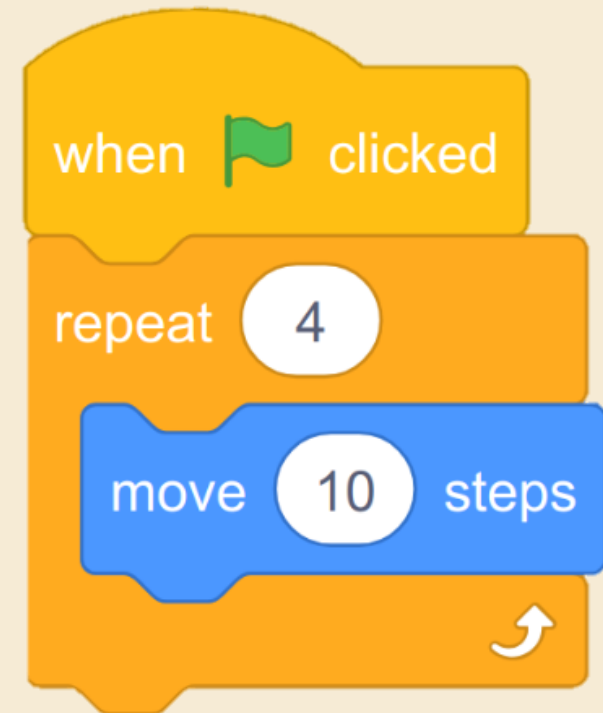
Programming Languages



Programming languages can translate between human and computer language.



programming languages



block-based coding

Building Blossom's Head



Building Blossom's Head

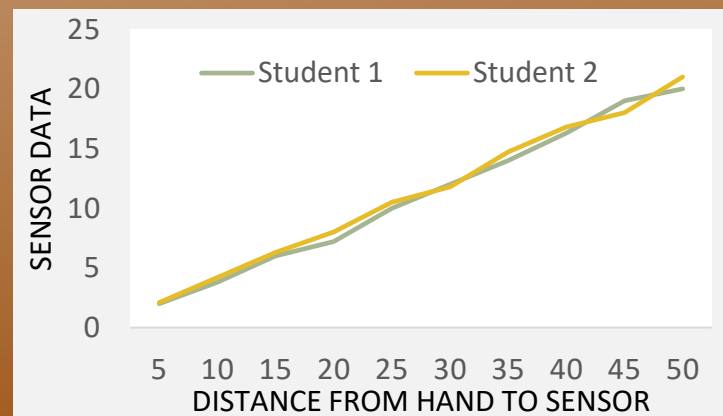




Extension: Sensor Input Data

- let students go to the sensor input page on Blossom's interface
- let students inspect the data from the ultrasonic sensor and how it changes as they move their hand towards it or further away
- sensor data to diagram
 - let students measure the distance from their hand to the sensor with a ruler and plot it in a diagram (x-axis: distance, y-axis: the number of the sensor input data on the web interface)

Distance from hand to sensor in cm	5	10	20	50
Sensor input data	X (let students measure)	~ 2x	~ 4x	~ 10x



Questionnaire



- same questions as last week, fill out as honestly as you can

Summary – From Signals to Data



- What are sensors?
 - Parts of machines that help them sense things around them (take in information from the environment).
- Types of sensors
 - A lot of different sensors exist, let students name a few (e.g., temperature sensor, camera, microphone,

Teacher-only slide – Lesson Overview

Lesson 2 - Student success criteria



- I can explain what sensors are used for inside of robots and other machines.
- I can name three different types of sensors and what they measure.
- I can outline the order of processing inside the Blossom robot that is needed to turn environmental input into an action. (input → sensor → computer with code → motor position → movement)
- I can describe how computers use electricity to communicate information with off- and on-states (0/1).
- I can tell what a bit and byte are.
- I can encode and decode a short word in binary using the class chart.
- I can explain that a programming languages are used to translate between human (natural) language and computer (binary) language.

Summary – Sensing the World



- What are sensors?
 - parts of robots (and other machines) that notice things in the world around them
- Sensor types
- Control Process in Blossom: From environmental input (e.g. approaching hand) to movement
 - hand -> ultrasonic sensor
- How do computers communicate
- What is a bit, what is a byte?
- What are programming languages?